Based on the CSV file provided, the Amazon co-purchasing network contains **1,000 nodes** and **2,178 edges**.

1. Top 5 Nodes by Degree (Most Connections) These are the products most frequently bought with other items in this sample.

- **Node 458358** (Degree: 324)
- **Node 38365** (Degree: 151)
- **Node 175102** (Degree: 92)
- **Node 37981** (Degree: 46)
- **Node 498702** (Degree: 46)

2. Top 5 Nodes by Betweenness Centrality These products act as critical "bridges" between different clusters of products.

- **Node 458358** (Score: 0.472)
- **Node 58064** (Score: 0.444)
- **Node 24139** (Score: 0.426)
- **Node 522711** (Score: 0.340)
- **Node 38365** (Score: 0.204)

3. Top 5 Nodes by PageRank These products are highly influential because they are frequently co-purchased with other highly influential products.

- **Node 458358** (Score: 0.069)
- **Node 38365** (Score: 0.033)
- **Node 175102** (Score: 0.019)
- **Node 521221** (Score: 0.009)
- **Node 498702** (Score: 0.008)

Social and Community Structures in Networks

The network of products purchased together via Amazon has a small number of very highly connected “hub” products and a relatively large number of distinct clusters of products. Using the Modularity algorithm, we identified a number of distinct communities in the network. In an e-commerce environment, these communities may well correspond to particular categories or niches (for example, a cluster of digital cameras purchased as a group with SD cards and camera bags).

Influential Nodes

Node 458358 is the clear center of this sample of the network based on its ranking across all three centrality measures (Degree, Betweenness, and PageRank). Because of the large Degree (324), Node 458358 is a highly successful core product. Due to the high value of Betweenness Centrality (0.472), this node serves as a vital link connecting otherwise disparate product categories. Therefore, if Amazon were to run a cross-category promotional campaign, Node 458358 would be the ideal product to include in that promotion because it connects purchasers from different communities across different categories.

In contrast, nodes 58064 and 24139 have very high values for Betweenness Centrality, but do not even rank in the top 5 for Degree. This illustrates a key property of the network: a node doesn't need to have a large number of direct connections to be influential; it just needs to be located on the shortest path between distinctly different locations in the network and therefore constitutes an important recommendation link between different consumer behavior trends.